

Exercise 1: Random Vectors

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Problem 1 (LOTUS):

Recall the law of the unconscious statistician (LOTUS) is a theorem used to calculate the expected value of a function $g(X)$ of a random variable X that states that

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) dF_X(x) \quad (1)$$

Calculate the expectation of $g(x) = \log x^2$ in two different ways: 1) using a change of variable, and 2) using LOTUS.

Problem 2 (Rotation invariance):

Consider an orthogonal matrix $\mathbf{O} \in \mathbb{R}^{d \times d}$ and a random vector $\mathbf{v} \sim \mathcal{N}(0, \mathbf{I}_d)$. Show that

$$\mathbf{g} := \mathbf{O}\mathbf{v} \sim \mathcal{N}(0, \mathbf{I}_d),$$

i.e. the coordinates g_i are independent standard normal random variables $\mathcal{N}(0, 1)$.

Problem 3 (Reduction to isotropy):

- i) Let $\mathbf{z}$ be an isotropic mean-zero random vector in $\mathbb{R}^d$. Let $\boldsymbol{\mu} \in \mathbb{R}^d$ be a fixed vector and $\boldsymbol{\Sigma}$ be a fixed $d \times d$ positive-semidefinite matrix. Check that the random vector

$$\mathbf{x} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \mathbf{z}$$

has mean $\boldsymbol{\mu}$ and covariance matrix $\text{Cov}(\mathbf{x}) = \boldsymbol{\Sigma}$.

- ii) Let $\mathbf{x}$ be a random vector with invertible covariance matrix $\boldsymbol{\Sigma}$. Check that the random vector

$$\mathbf{z} := \boldsymbol{\Sigma}^{-1/2}(\mathbf{x} - \boldsymbol{\mu})$$

is an isotropic mean-zero random vector.

Problem 4 (Quadratic forms):

Suppose $\mathbf{y} \in \mathbb{R}^d$ is a random vector such that $\mathbb{E}\mathbf{y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{y}) = \boldsymbol{\Sigma}$.

- For any fixed matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, show that

$$\mathbb{E}[\mathbf{y}^\top \mathbf{A} \mathbf{y}] = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \mathbb{E}[(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{A} (\mathbf{y} - \boldsymbol{\mu})].$$

- Then, show that

$$\mathbb{E}[\mathbf{y}^\top \mathbf{A} \mathbf{y}] = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \text{tr}(\mathbf{A} \boldsymbol{\Sigma}). \quad (2)$$